Subject Description Form

Subject Code	ITC519
Subject Title	Information Technology for Textiles & Clothing Industries
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	This subject aims to provide students with a solid comprehensive knowledge of the theories and application of the state-of-art information technologies used in the textiles and clothing industries. At the completion of this subject, the student will be able to: 1. implement information technology in their related work; and 2. evaluate the capabilities and limitations of the application for each state-of-art information technology.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: 1. Understand and explain the role of value of information technology in the fashion and textiles industries. 2. Set up and evaluate appropriate policies to handle the operation of information technologies in fashion merchandising and retailing companies. 3. Use their knowledge of information technology, its capabilities and limitations, to carry out the related procurement projects. 4. Develop new business models based on newly emerging information technologies.
Subject Synopsis/ Indicative Syllabus	1. <u>Overview of Information Technology Development in Textiles and Clothing Industries</u> Review of the state-of-art information technology, the necessity and its future trend in the industry. Hardware and software developments in textiles and clothing industries. 2. <u>Telecommunications and Networking</u> The concept of telecommunication and its current development. Networking technology, different types of network, internet, intranet and extranet. Applications in textiles and clothing industries such as smart mirror and fitting room using RFID, real-time data capture, real-time inventory control.

	3. Managing Data Resources in Textiles and Clothing Industries Data management in today's fashion business. Data modelling, its related structure and database systems in textiles and clothing industries. Overview of data warehouse and advanced data analysis techniques including OLAP and data mining. 4. E-commerce Application of B2B and B2C E-commerce in fashion industry. E-commerce implementation strategies and pitfalls to avoid 5. Information System (IS) and Decision Supports Role of IS in an organization for problem solving, type of IS and its features. Introduction to various enterprise information systems including enterprise resources planning (ERP), product lifecycle management (PLM), supply chain management (SCM) and customer relationship management (CRM) systems and the related issues. 6. Knowledge Management in Contemporary Business Operations Application of information technology to support decision making in textiles and clothing industry. The contemporary issue of knowledge discovery and knowledge management in enterprises, and its applications in textiles and clothing industries. 7. Developing and Managing Information Systems The reasons to have system development. Different approaches and methodologies of information system development, and its pros and cons. 8. Security and Policy Review the needs of security policy in information processing. Different strategies to handle social networking. 9. Big Data and Deep Learning Introduce the idea of big data and deep learning, and review the existing application of AI in the fashion industry. 10. 3D Printing Introduce the idea of 3D printing and their existing application in shoes and apparel. The information flow model related to 3D printing and 3D CAD design and Internet business. 11. Introduction of Emerging Technology Introduce the pros and cons of the emerging technology, such as Clouding Computing, Unified Communication, Knowledge Engineering,
Teaching/Learning Methodology	It provides a comprehensive introduction of the information technology used in the textiles and clothing industries. Theory and practice are both included in this subject. To enhance the understanding of how information technology is actually implemented, the state-of-art computer software used in textiles and clothing industries will be demonstrated during the workshop. Lectures will be supplemented with

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<td>video sessions to enhance student's learning. Student will be expected to understand the theories of the state-of-art information technology and how to implement in the textiles and clothing industries through the assignments and mini-projects.</td>
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<tr>
<td>Assessment Methods in Alignment with Intended Learning Outcomes</td>
<td>
<table>
<tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="4">Intended subject learning outcomes to be assessed</td></tr>
<tr><td>1</td><td>2</td><td>3</td><td>4</td></tr>
<tr><td>1. Coursework
a) Assignments and mini- projects
b) Test
2. Examination</td><td>60%

40%
0%</td><td>✓

✓</td><td>

✓</td><td>✓

✓</td><td>✓

✓</td></tr>
<tr><td>Total</td><td>100%</td><td colspan="4"></td></tr>
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Group Mini-Projects allow the students to work in a group and discuss on the theory and application of the information technology. The individual test requires the students to assess their ability and knowledge to adopt information technology in solving problems. There is no restriction of using GenAI for all coursework but students should address the citation in all coursework.
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<td>Student Study Effort Expected</td>
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<tr><td>Class contact:</td><td></td></tr>
<tr><td>▪ Lectures</td><td>27 Hrs.</td></tr>
<tr><td>▪ Tutorials</td><td>6 Hrs.</td></tr>
<tr><td>▪ Workshops</td><td>6 Hrs.</td></tr>
<tr><td>Other student study effort:</td><td></td></tr>
<tr><td>▪ Student Self-Study</td><td>31 Hrs.</td></tr>
<tr><td>▪ Assignment/mini-project</td><td>35 Hrs.</td></tr>
<tr><td>Total student study effort</td><td>105 Hrs.</td></tr>
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<tr>
<td>Reading List and References</td>
<td><u>Books</u>
1. Date, C.J., An Introduction to Database Systems, Latest Edition, Addison Wesley.

2. Jackson, T., Neural Computing: An Introduction, Institute of Physics Publishing, Latest Edition.

3. Long, L. Introduction to Computers and Information Processing, Latest Edition, Prentice Hall.</td>
</tr>
</table>

	4. McLeod, R. Management Information Systems, Latest Edition, Prentice Hall. 5. Turban, E., Electronic Commerce: a managerial perspective, Prentice-Hall, New Jersey, Latest Edition. <u>Recommended Reading List</u> 1. Aldrich, W. (ed), CAD in Clothing and Textiles: A collection of expert views, Latest Edition, Blackwell Science, 1994. 2. Anderson, David L., Management Information Systems: Using Cases Within An Industry Context to Solve Business Problems with Information Technology, Upper Saddle River, J.J.: Prentice Hall, 2000. 3. Cashin, J., E-commerce Success: Building a Global Business Architecture, Computer Technology Research, 1999. 4. Chester, M., Electronic commerce and business communications, Springer, London, 1998. 5. Dilworth, J.B., Operations Management: Providing Value in Goods and Services, Latest Edition, Orlando, Fla, Dryden, 2000. 6. Evans, J.R. and Olson, D.L., Introduction to Simulation and Risk Analysis, Latest Edition, Upper Saddle River, N.J.: Prentice Hall, 2002. 7. Hunter, A., King, R. and Lowson, R.H., The Textile/Clothing Pipeline and Quick Response Management, Manchester, England: Textile Institute, 2002. 8. Post, Gerald V. and Anderson, David L., Management Information Systems: Solving Business Problems with Information Technology, Latest Edition, Boston: McGraw-Hill/Irwin, 2003. 9. Stair, R.M. and Reynolds, G.W., Principles of Information Systems: A Managerial Approach, Latest Edition, B&F Publishing Co., 2003. 10. Watson, R.T., Electronic Commerce: The Strategic Perspective, Dryden Press, 2000. 11. Budmen, I. and Rotolo, A. The book on 3D printing, CreateSpace, 2013. 12. Michael, M., Big data, big analytics: emerging business intelligence and analytic trends for today's businesses, Hoboken, New Jersey: John Wiley & Sons, Inc. 2013.